CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 27: Graphical Solution

Topic-wise short notes with solved examples

Solving a pair of simultaneous linear equations by drawing their graphs, recognising when the lines intersect, are parallel or coincide, and finding areas enclosed by lines.

1 The idea

- The graph of a linear equation in x and y is a straight line. Each point on it is a solution of the equation.
- A pair of equations is solved by drawing **both lines on the same axes**. A point that lies on both lines satisfies both equations.
- **The solution is the point of intersection of the two lines.**

2 Steps

- Write each equation with y (or x) as the subject.
- For each equation make a table of **three** points. Choose values that give whole numbers where possible.
- Choose a suitable scale, the same on both axes unless told otherwise, and state it.
- Plot the points and draw each line with a ruler. Label each line with its equation.
- Read the co-ordinates of the point of intersection. Check by substituting in both equations.

Example 2.1 Solve graphically: $x + y = 5$ and $2x - y = 4$.

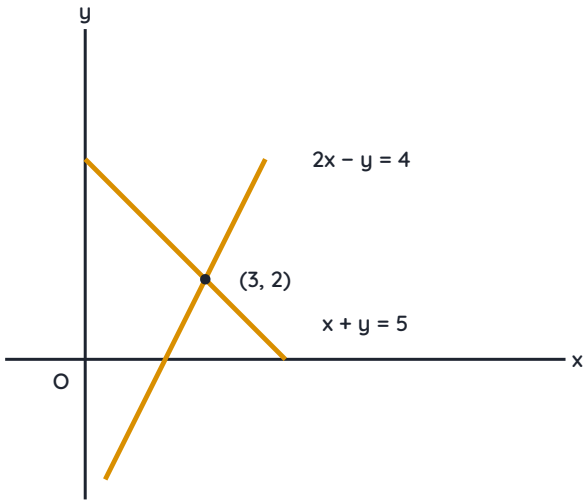
SOLUTION

For $x + y = 5$: points $(0, 5)$, $(2, 3)$, $(5, 0)$.

For $2x - y = 4$: points $(0, -4)$, $(2, 0)$, $(4, 4)$.

The lines cross at $(3, 2)$. Check: $3 + 2 = 5$ and $6 - 2 = 4$.

$x = 3, y = 2$



The two lines meet at (3, 2)

Example 2.2 Solve graphically: $3x + 2y = 12$ and $x - y = -1$.

SOLUTION

$3x + 2y = 12$: $(0, 6)$, $(2, 3)$, $(4, 0)$. $x - y = -1$: $(0, 1)$, $(1, 2)$, $(3, 4)$.

Both lines pass through $(2, 3)$.

$x = 2, y = 3$

3 Three possibilities

- For the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$:

Graph	Ratios	Solutions	Name
Lines intersect at one point	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Exactly one	Consistent
Lines are parallel	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	None	Inconsistent
Lines coincide	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Infinitely many	Dependent (consistent)

Example 3.1 Without drawing, say how the lines $2x + 3y = 7$ and $4x + 6y = 9$ are related.

SOLUTION

$\frac{2}{4} = \frac{3}{6} = \frac{1}{2}$, but $\frac{7}{9} \neq \frac{1}{2}$

Parallel; no solution

Example 3.2 How are $x - 2y = 3$ and $3x - 6y = 9$ related?

SOLUTION

$\frac{1}{3} = \frac{-2}{-6} = \frac{3}{9}$

The same line; infinitely many solutions

Example 3.3 For what value of k do $kx + 3y = 6$ and $2x + y = 5$ have no solution?

SOLUTION

$\frac{k}{2} = \frac{3}{1}$ gives $k = 6$. Then $\frac{6}{5} \neq 3$, so the lines are parallel.

$k = 6$

4 Lines parallel to the axes

- $x = a$ is a vertical line and $y = b$ is a horizontal line. They meet at (a, b).
- The line $y = x$ passes through the origin at 45° to the axes.

Example 4.1 Find graphically the point where $x = 3$ meets $2x + y = 10$.

SOLUTION

On $x = 3$: $y = 10 - 6 = 4$

$(3, 4)$

5 Triangles formed by lines

- To find the triangle formed by two lines and an axis, find the **three vertices**: the point of intersection of the lines, and the points where each line meets that axis.
- With the **x-axis**: put $y = 0$ in each equation. With the **y-axis**: put $x = 0$.
- Area $= \frac{1}{2} \times$ base (along the axis) $\times$ height (distance of the third vertex from that axis).

Example 5.1 The lines $x + y = 5$ and $2x - y = 4$ meet at $(3, 2)$. Find the area of the triangle they form with the x-axis.

SOLUTION

$x + y = 5$ meets the x-axis at $(5, 0)$; $2x - y = 4$ meets it at $(2, 0)$. Base $= 3$, height $= 2$.

3 square units

Example 5.2 Find the area of the triangle formed by the same two lines and the y-axis.

SOLUTION

They meet the y-axis at $(0, 5)$ and $(0, -4)$. Base $= 9$, height $= 3$ (the x co-ordinate of the intersection).

13.5 square units

Example 5.3 Find the area of the region bounded by $x = 0$, $y = 0$, $x = 4$ and $y = 3$.

SOLUTION

The region is a rectangle 4 by 3.

12 square units

Example 5.4 The lines $y = x$, $y = 2x$ and $x = 3$ form a triangle. Find its vertices and area.

SOLUTION

$y = x$ and $y = 2x$ meet at $(0, 0)$. On $x = 3$: the points are $(3, 3)$ and $(3, 6)$.

Base along $x = 3$ is 3; height from the origin is 3.

Vertices $(0, 0)$, $(3, 3)$, $(3, 6)$; area 4.5 square units

6 Word problems

- Form two equations, draw both graphs and read the point of intersection. Only the part of the graph that makes sense (for example, positive values) is used.

Example 6.1 The sum of two numbers is 8 and their difference is 2. Find them graphically.

SOLUTION

$x + y = 8$: $(0, 8)$, $(4, 4)$, $(8, 0)$. $x - y = 2$: $(2, 0)$, $(4, 2)$, $(6, 4)$.

The lines meet at $(5, 3)$.

5 and 3

7 Common mistakes

- Plotting only two points for a line, so that an error goes unnoticed.
- Using different scales on the two axes without saying so.
- Reading the intersection inaccurately because the lines were not drawn with a ruler or were too short.
- Giving the answer as a point when the values of x and y are asked, or the other way round.
- For the area with the y-axis, using the y co-ordinate of the intersection as the height. The height is its x co-ordinate.
- Concluding "no solution" when the lines merely do not meet on the part of the paper drawn. Extend them.